Lab 1 — Exercise: Notebooks & Markdown

ITCS 3162 — Introduction to Data Mining

Name: *your name here* **Date:** *today's date*

Complete the exercises below. For each task:

- **Code tasks** → fill in the code cell
- **Written answers** → edit the markdown cell directly (look for `YOUR ANSWER:` prompts)

When you're done, run **Kernel → Restart & Run All** to make sure the whole notebook executes top-to-bottom without errors, then download the `.ipynb` and submit via Canvas.

Exercise 1 — Running cells (5 pts)

Run the cell below. Then **modify** it so that it prints your full name and your major.

```
# TODO: change the strings to your name and major
name = "Srinivas Dengle"
major = "Computer Science"
print(f"Hi, I'm {name} and I'm studying {major}.")
```

```
Hi, I'm Srinivas Dengle and I'm studying Computer Science.
```

Exercise 2 — Variable persistence (10 pts)

Write code in the next cell that creates two variables, `a` and `b`, with any numeric values you like. Then in the cell after that, print their sum, difference, and product on three separate lines.

```
# TODO: define a and b
a=12
b=5
```

```
# TODO: print a+b, a-b, a*b on separate lines
print(a+b)
print(a-b)
print(a*b)
```

```
17
7
60
```

Exercise 3 — Markdown formatting (15 pts)

Double-click this cell and replace the placeholder content below with:

1. A level-2 heading with the text "About me"
2. A bulleted list with three items: your favorite class, a hobby, and a goal for this course
3. One sentence containing at least one **bold** and one *italic* word
4. An inline code reference to `pandas`
5. A link to a website you find useful

YOUR ANSWER (replace this line with your formatted markdown):

1. About me:

My name is Srinivas Dengle, pursing computer science and currently a senior.

2.

Favorite class: 3162 Data Mining Hobby: Hiking, Gym and soccer Goal: Learn skills required for Data Mining.

3.

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4. `pd.Dataframe()`

5. https://www.w3schools.com/python/pandas/default.asp

Exercise 4 — Math notation (10 pts)

Replace this cell's content with the LaTeX rendering of the **slope-intercept form** of a line, displayed (not inline). Then below it, write one sentence using inline math to explain what `m` represents.

YOUR ANSWER:

[y = mx + b]

In the equation (y = mx + b), the variable (m) represents the slope of the line.

Exercise 5 — A small table (10 pts)

In the cell below, create a markdown table listing **three data mining tasks** we'll cover this semester. Columns: `Task`, `Type` (supervised / unsupervised / association), and `Example use case`.

YOUR ANSWER:

Task	Type	Example use case
Classification	Supervised	Predicting whether an email is spam
Clustering	Unsupervised	Grouping customers by shopping behavior
Market Basket Analysis	Association	Finding products frequently bought together

Exercise 6 — Reflection (10 pts)

In 2–4 sentences, describe one thing you found confusing about notebooks and one thing you find useful compared to writing a regular Python script.

YOUR ANSWER: One thing that was confusing was creating a table in the markdown cell.
Benefit was its ability to run regular code in blocks and see immediate output without executing the entire program.

Submission checklist

- [ ] Replaced your name and date at the top
- [ ] All code cells run without errors
- [ ] All `YOUR ANSWER` prompts have been replaced
- [ ] Ran **Restart & Run All** before exporting
- [ ] Downloaded as `.ipynb` and uploaded to Canvas

```
Start coding or generate with AI.
```